WORKED SOLUTION

STEP II 2010, Question 10

Mechanics • Elastic Collisions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q10

This question has a full guided treatment in *Beyond STEP*, Section 10.2, *The Stacked-Ball Effect*: https://beyondstep.keithmcnulty.org/ch10/stacked-balls.html

The key idea

Everything rests on one collision: a particle hitting a stationary particle elastically. Conservation of momentum plus Newton's experimental law (with $e = 1$) give the speed passed on, and it depends only on the *ratio* of the masses. In part (ii) every collision has the same mass ratio λ, so each one multiplies the speed by the same factor, and a geometric progression does the rest.

Solution

Model the particles as point masses moving along a straight line on a smooth table, with all velocities measured in the direction of B's initial motion.

Part (i)

Let B and A have velocities v_B and v_A just after the collision. Conservation of momentum gives

$$bmu = bm\,v_B + m\,v_A,$$

and because the collision is perfectly elastic ($e = 1$), Newton's experimental law gives speed of separation equal to speed of approach:

$$v_A - v_B = u.$$

Substituting $v_B = v_A - u$ into the momentum equation and cancelling m,

$$bu = b(v_A - u) + v_A \quad \Longrightarrow \quad v_A = \frac{2bu}{b+1}.$$

So the speed of A is $\dfrac{2b}{b+1}u$.

This formula only used the ratio b of the masses, and the derivation works for any $b > 0$. Since $b+1 > b$ and $u > 0$,

$$v_A = 2u \cdot \frac{b}{b+1} < 2u,$$

whatever the value of b. So A can never be made to move at twice B's initial speed, though by taking b large it can get as close to $2u$ as we like.

Part (ii)

The masses, in order along the line, are

$$B_1 : \ \lambda^n m, \quad B_2 : \ \lambda^{n-1} m, \quad \ldots, \quad B_n : \ \lambda m, \quad A : \ m.$$

Each particle is λ times as heavy as the next one along. So whenever a particle moving with speed v hits the next (stationary) one, part (i) applies with $b = \lambda$, and the struck particle moves off with speed kv, where

$$k = \frac{2\lambda}{\lambda+1}.$$

From the equation $v_B = v_A - u$ in part (i), the striking particle carries on forwards with speed

$$kv - v = \frac{\lambda - 1}{\lambda + 1} v,$$

which is positive since $\lambda > 1$.

So B_1 gives B_2 speed ku, then B_2 gives B_3 speed $k^2 u$, and so on, until B_n, moving at $k^{n-1}u$, gives A speed

$$v_A = k^n u = \left(\frac{2\lambda}{\lambda+1}\right)^n u.$$

Ask yourself Could there be extra collisions, with a B_i catching up with the one in front after it has passed its speed on?

After all n collisions, B_i moves forward with speed $\dfrac{\lambda-1}{\lambda+1}k^{i-1}u$ for $i = 1, \ldots, n$, and A moves with speed $k^n u$. Since $\lambda > 1$ we have $k > 1$ (because $2\lambda > \lambda + 1$), so these speeds increase strictly along the line: each particle moves faster than the one behind it, and they separate. There are no further collisions, and A's final speed is $k^n u$.

Because $k > 1$, $k^n \to \infty$ as $n \to \infty$. So for any target speed V, choosing n large enough that $k^n u > V$ makes A move faster than V: there is no upper limit on A's speed.

The case $\lambda = 4$

Now $k = \frac{8}{5}$, and we need the least n with $\left(\frac{8}{5}\right)^n > 20$. Taking logarithms to base 10 (an increasing function, so the inequality is preserved),

$$n \log_{10} \tfrac{8}{5} > \log_{10} 20.$$

Writing $\frac{8}{5} = \frac{2^4}{10}$ and $20 = 2 \times 10$,

$$\log_{10} \tfrac{8}{5} = 4 \log_{10} 2 - 1 \approx 0.20412,$$
$$\log_{10} 20 = \log_{10} 2 + 1 \approx 1.30103.$$

Since $\log_{10} \frac{8}{5} > 0$ we may divide by it, giving

$$n > \frac{1.30103}{0.20412} \approx 6.37.$$

The least such integer is $n = 7$. (The margin is comfortable: 6.37 is far from an integer, so the approximation for $\log_{10} 2$ cannot change the answer.)

Checks

- In part (i), $b = 1$ gives $v_A = u$ and $v_B = 0$: equal masses swap velocities, as they should. As $b \to \infty$, $v_A \to 2u$.
- Kinetic energy: with $v_A = \frac{2b}{b+1}u$ and $v_B = \frac{b-1}{b+1}u$, we get $bv_B^2 + v_A^2 = \frac{b(b-1)^2+4b^2}{(b+1)^2}u^2 = bu^2$, so energy is conserved.
- Directly: $1.6^6 \approx 16.78 < 20$ and $1.6^7 \approx 26.84 > 20$, confirming $n = 7$. A step-by-step simulation of the chain with exact fractions also gave A's speed as exactly $k^n u$ after exactly n collisions.

Common slips

- Getting the masses the wrong way round: B_1 is the *heaviest* ($\lambda^n m$) and B_n the lightest of the Bs (λm), with A lighter still.
- Counting $n - 1$ collisions instead of n: the speed is passed along $B_1 \to B_2 \to \cdots \to B_n \to A$, which is n steps.
- Forgetting to rule out later collisions, or giving the non-integer 6.37 instead of the least integer $n = 7$.

Beyond STEP

A single elastic collision can at most double the speed, but a chain of collisions has no such limit. This is the idea behind the "stacked ball" demonstration, where a light ball

dropped on top of a heavy one rebounds far higher than it fell. Analysing collisions through momentum and energy (and, more neatly, in the centre-of-mass frame, where an elastic collision just reverses each velocity) is a central theme of first-year mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats it in depth.

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